

Name: _____

TOPIC TEST

Algebraic techniques

- Time allowed: 45 minutes
- Part A: 10 multiple-choice questions (10 marks)
- Part B: 10 free-response questions (40 marks)
- Total: 50 marks

Part A

10 multiple-choice questions

1 mark each: 10 marks

Circle the correct answer.

1 Solve $3x + 9 = 4 - 2(x + 5)$.

- A** $x = -2$
B $x = 2$
C $x = -3$
D $x = -15$

2 $\frac{1}{(x+y)^{-3}}$ is equivalent to:

- A** $\left(\frac{1}{x+y}\right)^3$
B $(x+y)^3$
C $x^3 + y^3$
D $\frac{1}{x^3 + y^3}$

3 Simplify $\frac{5k^{-2} \times (4k^3)^2}{4k^{-1}}$.

- A** $\frac{5}{k^5}$
B $5k^5$
C $10k^5$
D $20k^5$

4 Expand $(a - 4)(a^2 + 2a - 3)$.

- A** $a^3 + 2a^2 + 11a + 12$
B $a^3 + 2a^2 + 11a - 12$
C $a^3 - 2a^2 - 11a + 12$
D $a^3 - 2a^2 - 11a - 12$

5 Factorise $9x^2 + 3xy - 2y^2$.

- A** $(3x - 2y)(3x + y)$
B $(3x + 2y)(3x + y)$
C $(3x - 2y)(3x - y)$
D $(3x + 2y)(3x - y)$

6 Simplify $\frac{n+2}{n^2 - n - 6}$.

- A** $\frac{2}{n-3}$
B $\frac{1}{n-3}$
C $\frac{1}{n+3}$
D $n - 3$

- 7 Simplify $m + n + \frac{1}{m-n}$.
- A $\frac{m+n+1}{m-n}$
- B $\frac{m^2+n^2+1}{m-n}$
- C $\frac{m^2-n^2+1}{m-n}$
- D $\frac{m^2-2mn-n^2+1}{m-n}$
- 8 The conversion of degrees Celsius to degrees Fahrenheit is given by the formula $F = \frac{9}{5}C + 32$. Find F when $C = -10$.
- A -50
- B -14
- C 50
- D 14
- 9 Evaluate $-x^5 + 2x^3 - 3x^2 + 4x - 5$ when $x = -1$.
- A -6
- B -5
- C 1
- D 7
- 10 Find the value of n if $2\sqrt{180} - \frac{\sqrt{80}}{2} = \sqrt{n}$.
- A 500
- B 50
- C 10
- D 5

Part B

10 free-response questions

40 marks

Show your working where appropriate.

11 Solve $\frac{x+5}{2} - \frac{4-x}{3} = 1$.

[3 marks]

12 Simplify each expression.

a $\frac{4^{n+3} \times 2^{4-n}}{8^{n+1}}$

b $2\sqrt{5a^4} \times \sqrt{5} \times \sqrt[3]{a}$

[6 marks]

- 13 Simplify and express with positive indices: $\frac{2w^6h^3}{9h^5} \div \frac{4w^{-5}h^7}{(3w^{-2}h^3)^2}$

[4 marks]

- 14 Expand and simplify $(2a - 3)(2a + 3)(a - 1)$.

[2 marks]

- 15 Factorise each expression.

a $45x^3 - 5xy^2 - 18x^2y + 2y^3$

b $3p^3 + 30p^2 + 75p$

[6 marks]

16 Simplify each expression.

a $\frac{p+2}{3} - \frac{p-1}{4}$

b $\frac{2m+1}{m+2} + \frac{3m}{m^2+m-2}$

[6 marks]

17 Simplify $\frac{2x^2-5x-3}{x+3} \div \frac{2x-6}{6x+18}$.

[4 marks]

- 18 If $a = 2\sqrt{3}$, then evaluate $a^3 + (a + 1)^2$.

[3 marks]

- 19 Simplify $\frac{12\sqrt{10}}{4\sqrt{2}} + \sqrt{80}$.

[2 marks]

- 20 Simplify $\frac{2}{\sqrt{5}+1} - \frac{1}{2-\sqrt{3}}$, writing your answer with a rational denominator.

[4 marks]

This is the end of the test.
Use the rest of this page for extra working space.

Answers**1** C **2** B **3** D **4** C **5** D**6** B **7** C **8** D **9** B **10** A

11 $x = -\frac{1}{5}$

12 a 2^{7-2n}

b $10\sqrt[3]{a^7}$

13 $\frac{w^7}{2h^3}$

14 $4a^3 - 4a^2 - 9a + 9$

15 a $(5x - 2y)(3x + y)(3x - y)$

b $3p(p + 5)^2$

16 a $\frac{p+11}{12}$

b $\frac{2m^2 + 2m - 1}{(m+2)(m-1)}$

17 $3(2x + 1)$

18 $28\sqrt{3} + 13$

19 $7\sqrt{5}$

20 $\frac{2\sqrt{5} - 4\sqrt{3} - 10}{4}$